Side [SimTalk]

Sets the **Side** of the *Frame* on which the *Interface* designated by <Path> is located.

Type

Attribute

Syntax

```
<Path>.Side:string
```

Assignment Value

You can assign a value of data type *string*.

You can specify `"Top"`, `"Right"`, `"Bottom"`, `"Left"`, or `"Angle-dependent"`.

`"Angle-dependent"` takes the angle between the objects into account when determining the starting point or the end point of the *Connector*.

Example

```
interface.Side := "right"
```

See also

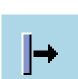
Side [drop-down list]

Source

Source

Use the object *Source* for producing the parts that move through your model. It usually represents the receiving department of your plant or the main machine producing the parts.

Image



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Description

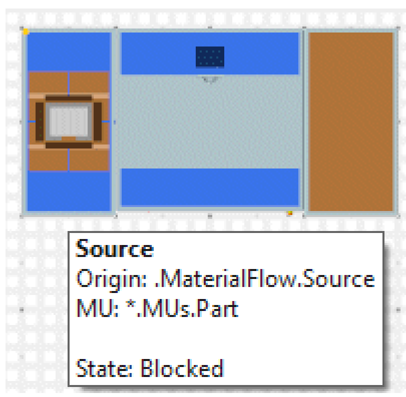
The *Source* has a capacity of one and no processing time. The *Source* produces the same or different types of *MUs* one after the other or in a mixed sequence. It might, for example, represent the *receiving* department of your plant that introduces parts produced at another location into the plant. Or it can be a machine, which produces the parts that the other stations process.

You can set a procedure to determine the times at which it creates the parts as well as a procedure to determine the types of *MUs* to be produced. As an active material flow object, the *Source* attempts to move the *MUs* it produced to the objects to which it is connected. You can also define how the *Source* proceeds when it cannot move a *MU* to its succeeding object, by selecting or clearing **Blocking** Operating Mode: ☑ Blocking .

You will use the *Source* to produce the parts and workpieces that move through your plant and which are processed by the different stations.

You will use the **Drain** to remove these parts from the factory to model, for example, the *shipping* department of your facility.

To show a **tooltip** with information about the *Source*, hover with the mouse over it.



To change the length of the graphic and the anchor points of the *Source*, click **Show Manipulators** on the **Edit** ribbon tab or press **M** on the keyboard.

Add the Object to the Simulation Model

To add the object *Source* to your simulation model, click **Manage Class Library** > **Basic Objects > MaterialFlow > Source** on the **Home** ribbon tab.

Compare the sample models: Click the **Window** ribbon tab, click **Start Page > Getting Started > Example Models > Small Examples**. Then, select the respective **Category**, the **Topic**, and the **Example** in the dialog **Examples Collection**, and click **Open Model**.

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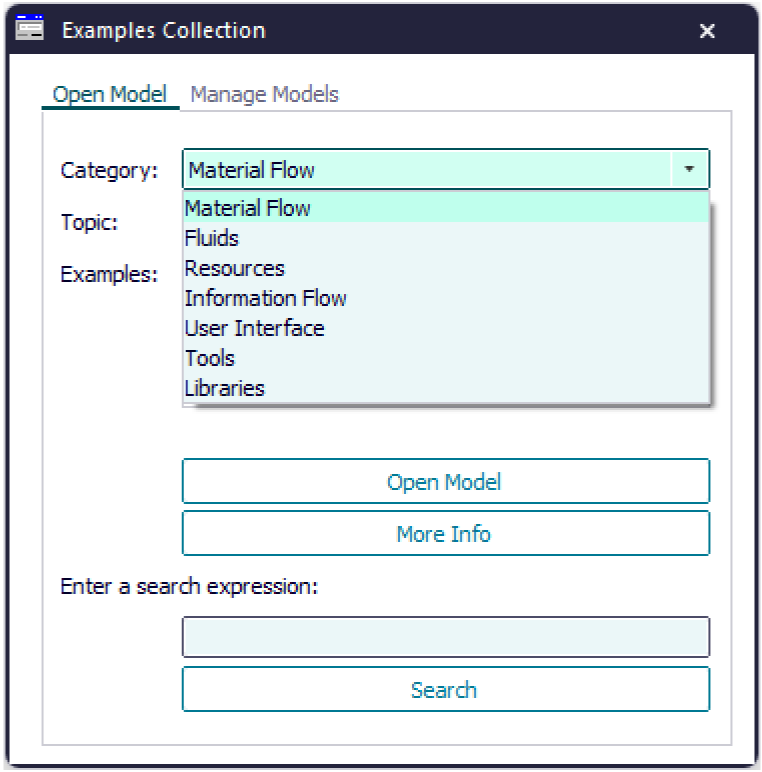

See also

Produce Parts with the Source

Check How Many Parts Were Introduced into the Plant

Dialog Box of the Source

Dialog Box of the Source

Double-click the icon of the *Source* to open its dialog box.

Edit Simulation Properties

In the dialog box you can change the **simulation properties** of the object. The shared properties are described under **Dialog Items of the Objects**.

Edit Animation Properties

To edit the **3D properties** of the object in the dialog box **Edit 3D Properties**:

- Click the button 

 Edit 3D Properties in the lower left corner of the simulation properties dialog box.
- Select the object in the model and press the **spacebar**.

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To manipulate the graphic of the object, click **Show Manipulators** on the **Edit** ribbon tab or press **M** on the keyboard.

Tab Attributes

Tab Attributes

The tab **Attributes** provides the settings, which the object offers.

The settings are listed in the table of contents to the left.

The shared properties are described under the **Tab Attributes**.

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Operating Mode [check box]

The **Operating Mode** sets how the *Source* proceeds if it cannot create the *MUs* at the **Time of Creation** you entered.

Remarks

You can:

- Select **Blocking**, to make the *Source* remember the time at which it was supposed to produce the next *MU*. It then produces the following *MUs* at the next possible point in time, i.e., when the *MU* that was blocked by the successor, was moved on. The *Source* stops creating parts when the simulation time has reached the **Stop** time.
- Clear **Blocking**, to make the *Source* create another *MU* exclusively at the time of creation you entered.

> **Note**
> When the *Source* is temporarily not operational because it is failed, paused or blocked, the times of creation may shift, if you select **Blocking**. Then, the settings for the times of creation cannot be realized.

SimTalk

[Blocking [SimTalk]]

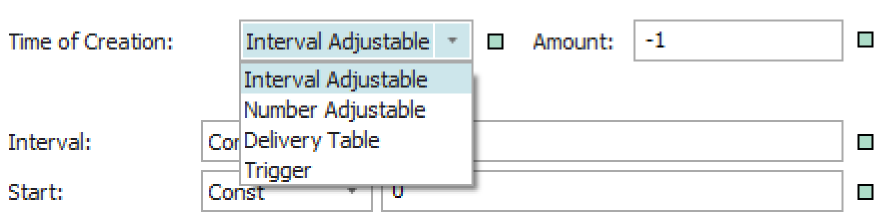
See also

[Time of Creation [Source]]

Time of Creation

Time of Creation [Source]

Select the point in time at which and how the *Source* produces *MUs* from the drop-down list.



Remarks

You can select one of these settings:

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- **Interval Adjustable**

  The *Source* produces the first *MU* at the **Start** time. The **Time of Creation** designates the **Interval** between two creation events. It produces the last *MU* at the **Stop** time.

  If you do not type in a **Stop** time for this setting, *Plant Simulation* produces the amount of parts you entered at the most. The default value **-1** designates an unlimited number of parts to be produced. Otherwise, the *Source* produces parts until the **Stop** time is reached and ignores the **Amount** of parts you entered.

  > **Note**
  >
  > If you select a **constant** time all *MUs* are produced at this point in time.

- **Number Adjustable**

  The *Source* produces the number of *MUs*, which you type in as the **Amount**. If you activate the check box **Generate as Batch**, the **Amount** does not designate the number of *parts*, but the number of *batches* that the *Source* produces. The times at which the *MUs* are created are distributed according to the distribution you select for the **Interval**.

  If you select a **constant** time all *MUs* are produced at this point in time.

- **Delivery Table**

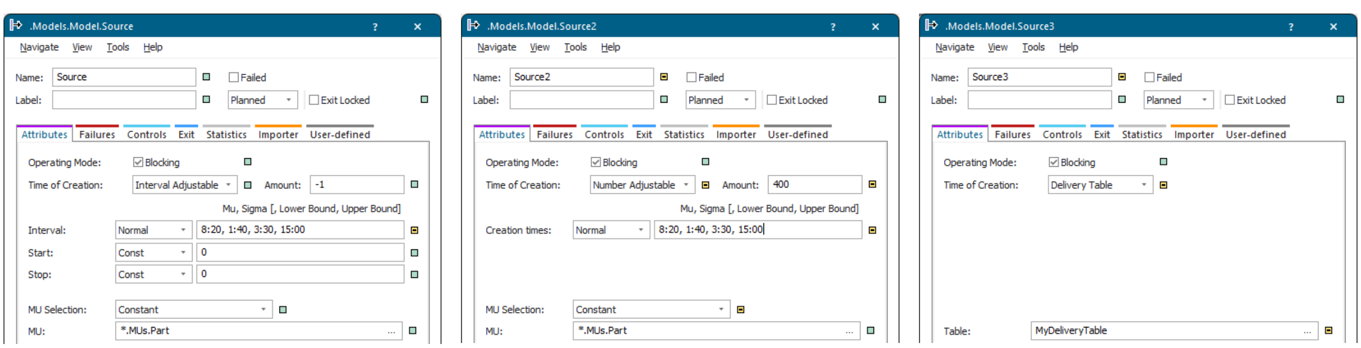
  The *Source* produces *MUs* according to the **Delivery Time**, the **Class**, the **Number**, the **Name**, and the **Attribute** which you type into the *Delivery Table*.

  > **Note**
  >
  > If you assign a new **Delivery Table** to the *Source* during the simulation, it also produces parts which were to be produced in the past according to the new delivery table.

- **Trigger**

  The *Source* produces *MUs* according to the values that a set of *Trigger* objects control.

  > **Note**
  >
  > Depending on the setting you select, the attribute **Path** (dialog item **MU** or **Table**) either sets the path to the *MU* class, the delivery list, the sequence table, the frequency table, or the percentage table.



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If the *Source* cannot produce the *MUs* at the **Time of Creation**, the **Operating Mode** determines how it proceeds:

- Select **Blocking** 

, to make the *Source* remember the time at which it was supposed to produce the next *MU*. It then produces the following *MUs* at the next possible point in time, i.e., when the *MU* that blocked the successor was moved on.
- Clear **Blocking**, to make the *Source* create another *MU* exclusively at the time of creation you typed in.

> **Note**
>
> When the *Source* is temporarily not operational because it is failed, paused or blocked, the **Time of Creation** may shift, if you select **Blocking**. Then, the settings for the times of creation cannot be realized.

SimTalk

TimeOfGeneration [SimTalk]

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Interval Adjustable [time of creation]

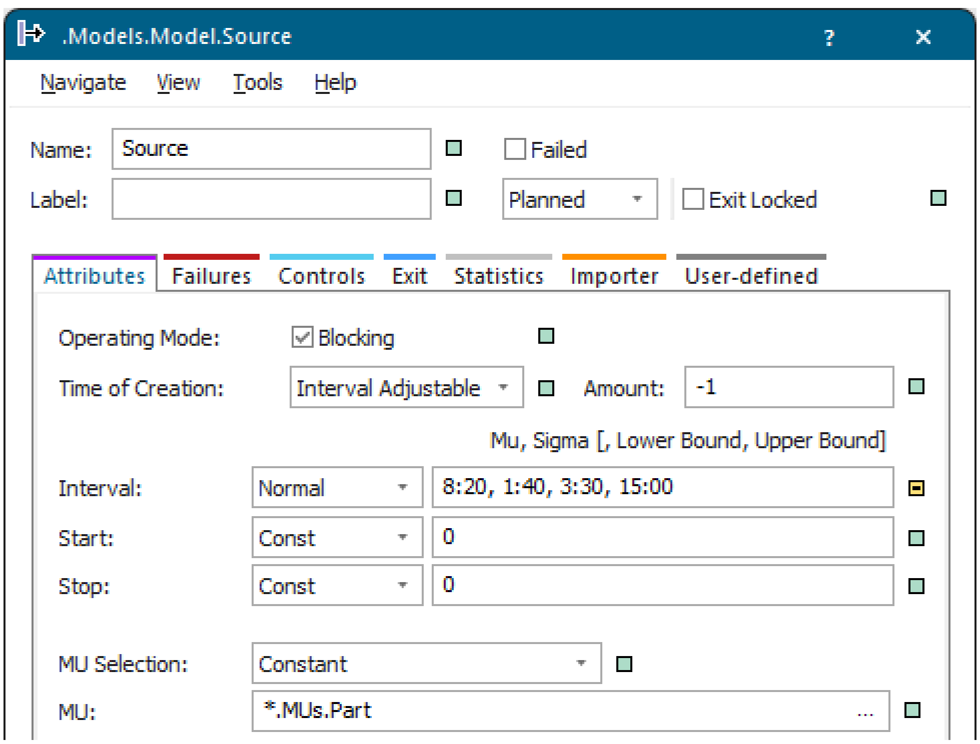
Produces the *MUs* according to point in time defined by the settings **Interval**, **Start**, and **Stop**.

Remarks

Select a distribution and enter the values, which that distribution requires, into the text box. *Plant Simulation* shows these values below the text box.

- The **Start** time designates the point in time at which the *Source* produces the first *MU*. It produces additional *MUs* after the time, which you enter as the **Interval**, has elapsed.
- The **Interval** designates the time span between two events at which the *Source* produces *MUs*.
- The **Stop** time designates the point in time at which the *Source* produces the last *MU*. If you type in `0` as the **Stop** time, the *Source* produces *MUs* according to the **Amount** you entered.

You can also select one of the **Probability Distributions** for the **Interval**, for the **Start** time, and for the **Stop** time.



Amount

Type in the amount of *MUs* which the *Source* produces.

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Note

The setting **Amount** is only available for the **Time of Creation > Number Adjustable** and **Interval Adjustable**.

If you do not type in a **Stop** time for the setting **Interval Adjustable**, *Plant Simulation* produces the amount of parts you typed in at the most. The default value of **-1** designates an unlimited number of parts to be produced. Otherwise, the *Source* produces parts until the **Stop** time is reached and ignores the **Amount** of parts you entered.

For the setting **Interval Adjustable** the *Source* might create more parts than the **Amount** you typed in, if the check box **Generate as Batch** is activated. This is because the batches are always produced in their entirety, not in part only.

For the setting **Time of Creation > Number Adjustable** the value does not designate the amount of *MUs*, but the amount of batches/lots which the *Source* produces if you activated **Generate as Batch**.

SimTalk

Number [SimTalk]

TimeOfGeneration [SimTalk]

See also

Produce Parts During an Interval Which You Define

Interval [Source]

Start [Source]

Stop [Source]

Interval [SimTalk] - Source

Start [SimTalk] - Source

Stop [SimTalk] - Source

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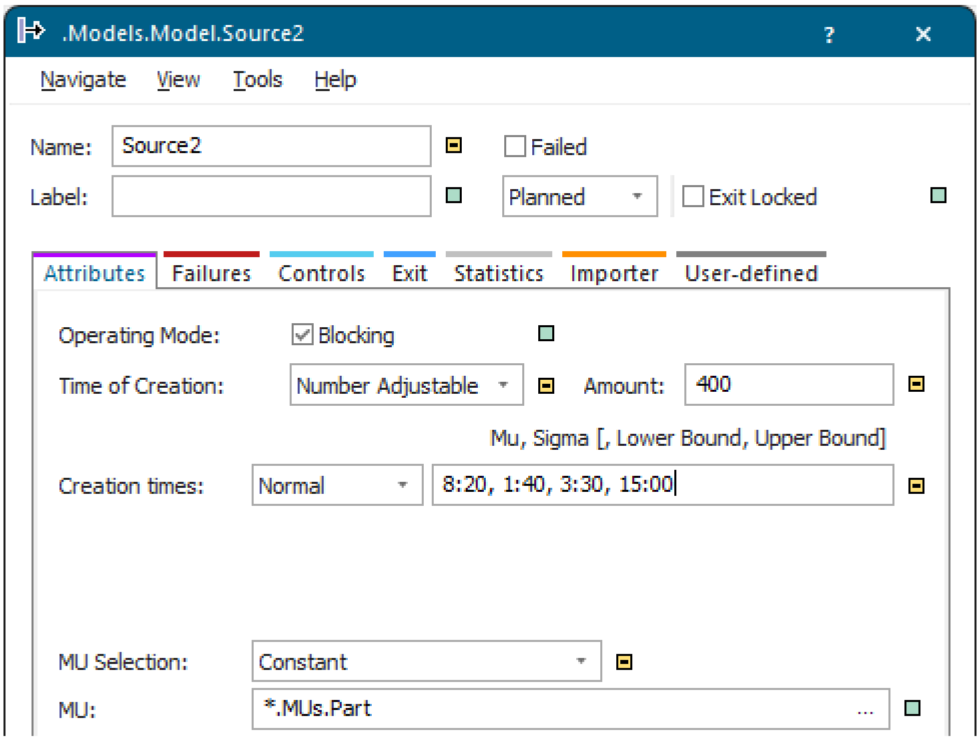
Number Adjustable [time of creation]

Produces the number of *MUs*, which you type into the text box **Amount** for the setting **Number Adjustable** of the *Source*.

Remarks

If you activate the check box **Generate as Batch**, the **Amount** does not designate the number of *parts*, but the number of *batches* that the *Source* produces.



You can also select one of the **Probability Distributions** from the drop-down list **Creation Times** and type the values that the selected distribution requires into the text box. The distribution determines the point in time at which the *Source* creates the *MUs*.

The *Source* produces the number of *MUs* at the different times which the random number generator generated at the beginning of the simulation. The determined times are contained within the bounds if you type in a lower bound and an upper bound.

If you select a **Constant** time, the *Source* creates the number of *MUs* all at one point in time.

Note

If you set the **Creation Times** via *SimTalk*, use the attribute **Interval**.

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> **Note**
> For the setting **Number Adjustable** you cannot select the **MU Selection** > **Sequence**.

SimTalk

Number [SimTalk]

TimeOfGeneration [SimTalk]

See also

Produce the Number of Parts You Need

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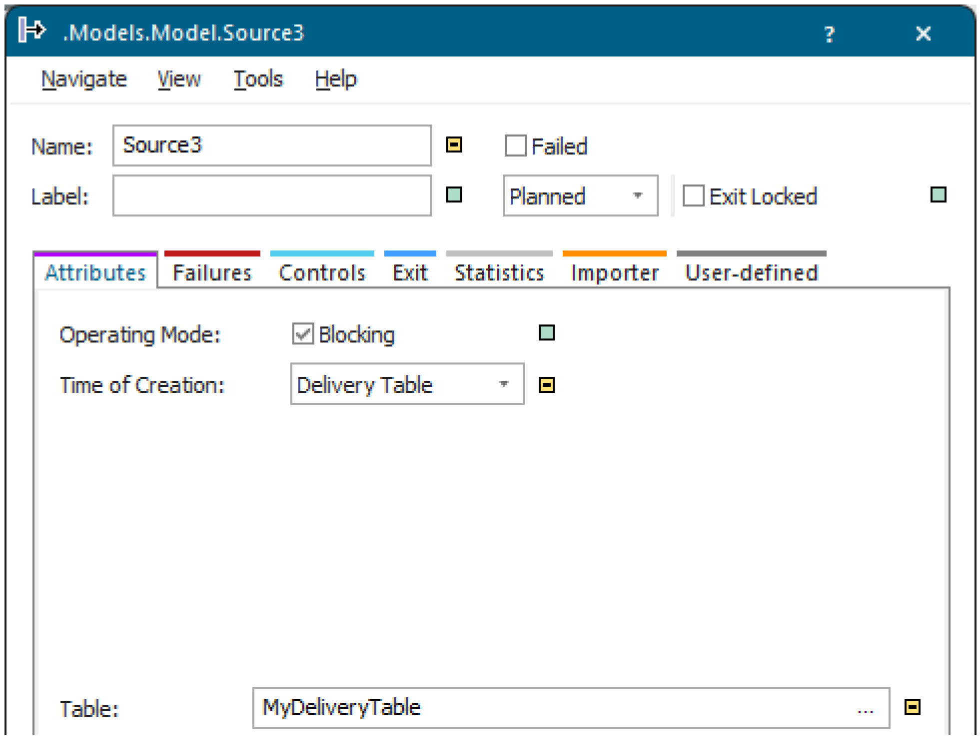
Delivery Table [time of creation]

Produces the *MUs* according to the settings, which you typed into the *Delivery Table* for the setting **Delivery Table** of the *Source*.



Remarks

Type the path of the **Delivery Table** into the text box next to **Table**.

Or click ... and select a table in the dialog **Select Object**.

You can also add information about the name and number of the produced parts as well as values for their attributes to the *Delivery Table*.

The *Delivery Table* has five columns with the data types *time*, *object*, *integer*, *string*, and *table*. Instead of the data type *time* you can also use the data types *date*, *dateTime*, or *real*. Each row of the *Delivery Table* defines one order for producing *MUs*:

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.Models.MyEnginePlant.MyDeliveryTable

	time 1	object 2	integer 3	string 4	table 5
string	Delivery Time	MU	Number	Name	Attributes
1	1:20.0000	.MUs.Part	1	box	properties
2	1:30.0000	.MUs.Part	3	screw	properties
3	1:40.0000	.MUs.Part	2	shaft	properties
4					

- Type the time at which the *Source* produces the *MUs* into the column **Delivery Time**.

> **Note**
>
> *Plant Simulation* processes the *Delivery Table* row by row. For this reason the time should increase from row to row. If *Plant Simulation* processes a row in which the time is located in the past, it creates the part immediately. This may not be what you would expect to happen.

- Type the class of the *MU* into the column **MU**. You can also drag the *MU* class into the respective cell and drop it there.
- Type the number of *MUs* to be produced into the column **Number**.

> **Note**
>
> If you enter `0` as **Number** into a row, *Plant Simulation* does take the specified interval for the next creating cycle into account. This means that *Plant Simulation* does not skip this row, but does not produce a part during this cycle.

- Enter a name for the *MUs* to be produced into the column **Name**.

> **Note**
>
> You have to type data in the columns **Delivery Time** and **MU**. If you do not type in a **Number**, *Plant Simulation* produces a single *MU*. If you do not type in a **Name**, the produced *MUs* inherit the name of their class.

- Type in the name of the table attributes and user-defined attributes that the *Source* sets when it produces the *MUs* into the column **Attributes**.

  Type the name of the attribute in column 1 of the attribute table. If an attribute with that name does not exist, *Plant Simulation* creates it. Set the data type of the attribute with the data type of an additional column in the attributes table.

> **Note**
>
> *Plant Simulation* assigns the data type of the column containing the value of the attribute to the data type of the attribute. The respective entry is the value of the generated attribute.
>
> If the data type of the column is *table/list/stack/queue*, *Plant Simulation* does not copy the sublists of the attribute table, but creates a reference to this list. The newly created *user-defined attribute* then

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accesses this list, i.e., when you change one of the two lists, the contents of the other list is changed as well.

Note

The *Source* does not produce parts whose time of creation is in the future when you assign a **Delivery Table** during the simulation.

Instead of using a *DataTable*, which you inserted into a *Frame*, you can also use a *user-defined attribute* of data type *table* of the *Source* as the **Delivery Table**.

SimTalk

Path [SimTalk] - Source

See also

creationTable [SimTalk] - Source

TimeOfGeneration [SimTalk]

Produce Parts According to a Delivery Table

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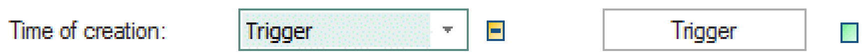
Trigger [time of creation]

Produces *MUs* according to the values that one or several *Trigger* objects control for the setting **Trigger** of the *Source*.



Remarks

Click **Trigger** and type the name(s) of the *Trigger* object(s) into the list that opens. Each *Trigger* in this list sends orders as *strings* to produce *MUs* to the *Source*.

Open the *Trigger*, click the tab **Values**, make sure that the **Trigger Type > Input** is selected and click **Values**.

Type the orders into the cells on the tab **Contents**:

- Type the **Point in Time** at which the *Source* creates *MUs* into the cells on the left-hand side.
- Type the **Value** of the order into the cells on the right-hand side. An order is a *string* with this format: Amount of *MUs* to be produced, type of the *MU* to be produced, type of the distribution, and the parameters of the distribution, (`<num>, <mu_Type>,<distributionType[, distribution parameters]>` in *SimTalk*).

  You always have to type in the amount of *MUs* to be produced, the type of the *MU* to be produced, and the distribution type. When you type in `Const` as the distribution type, the *Source* produces the *MUs* at the point in time which you type into the respective cell in the column **Point in Time**. If you would like it to produce the *MUs* with an offset to the time you entered there, type in number of seconds after which it produces them after the constant time (`Const`) you specified. For all other distribution types you have to enter the respective distribution parameters after the distribution type.

  If you select a distribution, its values set the time offset to the time you entered into the respective cell in the column **Point in Time**. The offset you type in has to be a positive number.

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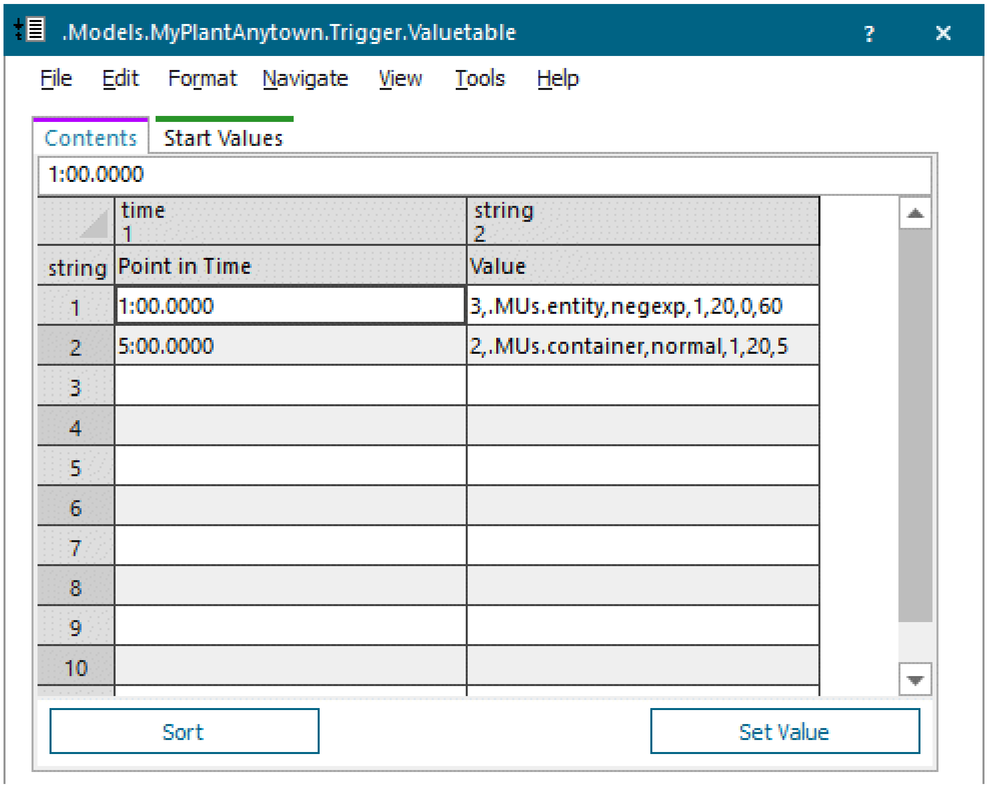

In the figure above the creation time for each *MU* is computed as `1:00.000 + z_negexp(…, 20, 0, 60)` and `5:00.000+z_normal(…, 20, 5)` respectively.

The random stream used is the internal stream of the *Source*.

If you type in the *Source* as a resource into the *ShiftCalendar*, *Plant Simulation* interrupts the production sequence during the weekends:

- If you select the **Operating Mode** > **Blocking**, the *Source* does not produce *MUs* at all during the weekends.
- If you clear **Blocking**, the *Source* produces and moves the *MUs*, as soon as its shift starts again.

You might, for example, use a *Source* that is controlled by a *Trigger*, if you want to set a time of creation, such as with a delivery table, and you would like to periodically repeat using the values in this list, for example when you select **MU Selection** > **Sequence Cyclical**.

SimTalk

Trigger [SimTalk]

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See also

Trigger [object]

Produce Parts Using a Trigger Object

creationTable [SimTalk] - Source

TimeOfGeneration [SimTalk]

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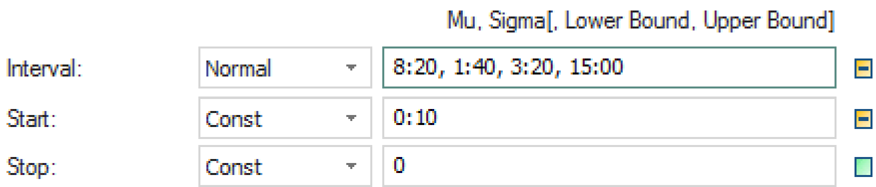
Interval [Source]

For **Time of Creation > Interval Adjustable** the **interval** is the time span that elapses between two events at which the *Source* produces parts. The *Source* creates an infinite number of *MUs*.



Remarks

Select a distribution from the drop-down list, and type the values, which that distribution requires, into the text box. *Plant Simulation* shows these values along the upper border of the tab.

You can also select **Const** and type in a constant time.

SimTalk

Interval [SimTalk] - Source

See also

Interval Adjustable [time of creation]

Start [SimTalk] - Source

Stop [SimTalk] - Source

Times and Distributions

Probability Distributions

Empirical Distributions

User-defined Distributions

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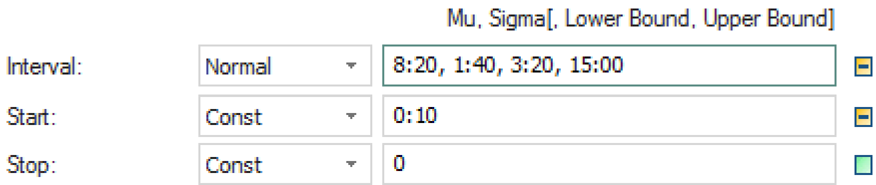
Start [Source]

The **Start** time designates the point in time at which the *Source* produces the first *MU*.



Remarks

Select a distribution from the drop-down list, and type the values, which that distribution requires, into the text box. *Plant Simulation* shows these values along the upper border of the tab. You can also type in a constant time (**Const**).

SimTalk

Start [SimTalk] - Source

See also

Interval [SimTalk] - Source

Stop [SimTalk] - Source

Times and Distributions

Probability Distributions

Empirical Distributions

User-defined Distributions

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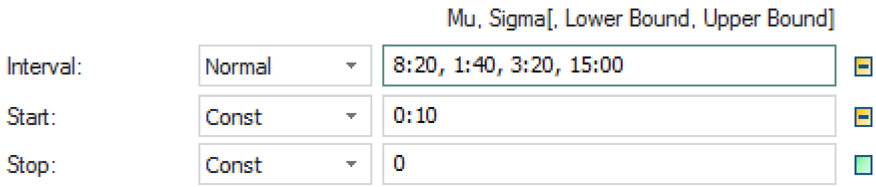
Stop [Source]

The **Stop** time designates the time at which the *Source* stops producing *MUs*.



Remarks

Select a distribution from the drop-down list, and type the values, which that distribution requires, into the text box. *Plant Simulation* shows these values along the upper border of the tab.

You can also type in a constant time (**Const**).

SimTalk

Stop [SimTalk] - Source [SimTalk]

See also

Interval [SimTalk] - Source

Start [SimTalk] - Source

Times and Distributions

Probability Distributions

Empirical Distributions

User-defined Distributions

MU Selection

MU selection [drop-down list]

Select which type of *MUs* and how the *Source* produces *MUs*.

Remarks

You can select one of these settings:

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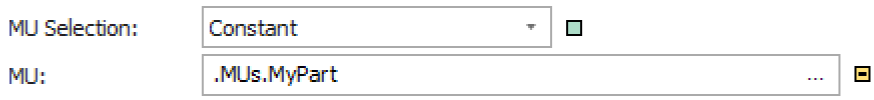
- **Constant**

  The *Source* produces one type of *MU* only. Type its path into the text box **MU**.



- **Sequence Cyclical**

  The *Source* produces the *MUs* in a fixed sequence, which you type into a table. As soon as the *Source* has processed the sequence, it repeats the production sequence periodically.

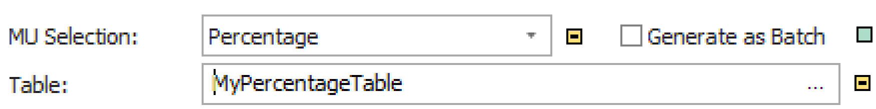
- **Sequence**

  The *Source* produces *MUs* according to the sequence table, whose name you typed into the text box next to **Table**. As opposed to **Sequence Cyclical**, **Sequence** processes the sequence only once, not repeatedly.

- **Random**

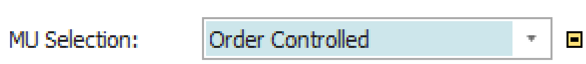
  The *Source* produces *MUs* in random frequencies according to the values which you typed into the frequency table.

- **Percentage**
- The *Source* produces *MUs* in percentages according to the values which you typed into the percentage table.



- **Order Controlled**

  The *Source* only produces *MUs* if they are ordered by a *Store* with activated setting **Supermarket** or with the method **orderParts**.



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> **Note**
>
> Depending on the **Time of Creation** you select, the dialog item **MU** or **Table** (attribute **Path**) either sets the path to the *MU* class, the delivery list, the sequence table, the frequency table, or the percentage table.

> **Note**
>
> To use a copy of the *DataTable* instead of a reference to the *DataTable*, assign the name `.unshare` to the subtable.

SimTalk

MUSelection [SimTalk]

Path [SimTalk] - Source

unshare [SimTalk]

See also

Store as **Supermarket [check box] Configuration [button]**

getCurrentOrderTableRow [SimTalk]

Video on YouTube

https://youtu.be/BvHLQCIjGIA?si=Jy0O3wvXpSNFltdg&t=100

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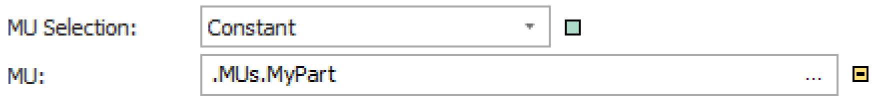
Constant [MU selection]

Produces one type of *MU* for the setting **MU Selection > Constant**.



Remarks

Type the path to this type of *MU* into the text box next to **MU**.

Or click ··· and select a *MU* class, **Part**, **Container** or **Transporter**, in the dialog **Select Object**.

Or select the *MU* in the *Toolbox* or the *Class Library*, drag it to the text box and drop it there.

SimTalk

MUSelection [SimTalk]

Path [SimTalk] - Source

See also

Produce a Single Part Type Only

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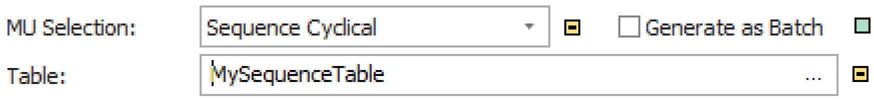
Sequence Cyclical [MU selection]

Produces parts according to the sequence table whose path you enter for the setting **MU Selection > Sequence Cyclical**.



Remarks

You can also click ··· and select the name of the table in the dialog **Select Object**. *Plant Simulation* automatically formats the table.

Or select the *MU* in the *Toolbox* or the *Class Library*, drag it to the text box and drop it there.

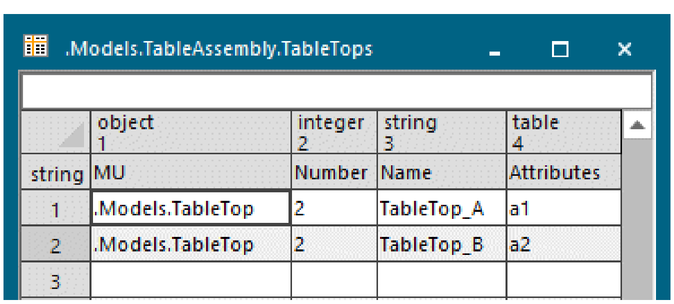
> **Note**
> For the *MUs* that are to be produced, you can also preallocate *user-defined attributes*. If the attribute has the data type *table*, the *MU* gets a reference to the table defined in the *Source*. A reference saves memory and increases performance. In some cases you might need a copy of the table though. You can enforce this by assigning the name `.unshare` to the subtable, compare **unshare**.

To make the *Source* produce the number of *MUs*, which you typed into the cell **Number** of the **Table**, in a single lot, select **Generate as Batch**.

If you select **Number Adjustable** as the **Time of Creation**, the **Amount** does not designate the number of *parts*, but the number of *batches*. Then the *Source* produces the entire set of parts all at once at the given start time and attempts to move all of the parts on to the next object in a single lot.

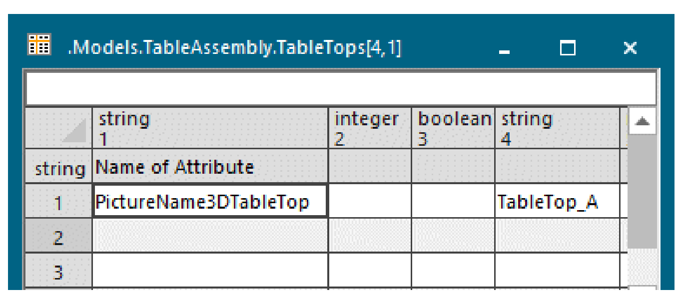
To make the *Source* produce the *MUs* as a sequence of individual parts one after the other, clear the check box.

Once the *Source* has processed the entire sequence, it starts processing the information in the table again, starting at the beginning of the sequence.



.Models.TableAssembly.TableTops

	object 1	integer 2	string 3	table 4
string	MU	Number	Name	Attributes
1	.Models.TableTop	2	TableTop_A	a1
2	.Models.TableTop	2	TableTop_B	a2
3				



.Models.TableAssembly.TableTops[4,1]

	string 1	integer 2	boolean 3	string 4
string	Name of Attribute			
1	PictureName3DTableTop			TableTop_A
2				
3				

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SimTalk

MUSelection [SimTalk]

Path [SimTalk] - Source

GenerateAsBatch [SimTalk]

getCurrentOrderTableRow [SimTalk]

unshare [SimTalk]

See also

Produce Parts in a Fixed Sequence Over and Over Again

Unpublished work. © 2026 Siemens

Sequence [MU selection]

Produces *MUs* according to the values which you typed into the sequence table, whose name you typed into the text box next to **Table** for the setting **MU Selection > Sequence**.

Remarks

You can also click ··· and select the name of the table in the dialog **Select Object**. *Plant Simulation* automatically formats the table.

Or select the *MU* in the *Toolbox* or the *Class Library*, drag it to the text box and drop it there.

Note

As opposed to the setting **Sequence Cyclical**, the setting **Sequence** processes the sequence only once, not repeatedly.

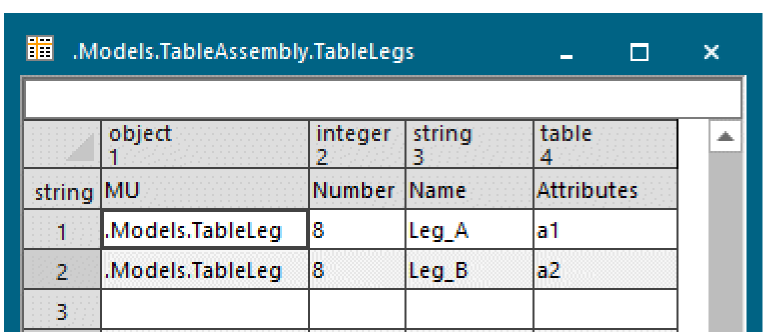
The setting **Sequence** is not available if you set **Time of Generation** to **Number Adjustable**.

Note

For the *MUs* to be produced you can also preallocate *user-defined attributes*. If the attribute has the data type *table*, the *MU* gets a reference to the table defined in the *Source*. A reference saves memory and increases performance. In some cases you might need a copy of the table though. You can enforce this by assigning the name `.unshare` to the subtable, compare **unshare**.

To make the *Source* produce the number of *MUs*, which you typed into the cell **Number** of the **Table** in a single lot, select **Generate as Batch** ☑ Generate as Batch. Then the *Source* produces the entire set of parts all at once at the given start time and attempts to move all of the parts on to the next object in a single lot.

To make the *Source* produce the *MUs* as a sequence of individual *MUs*, clear the check box.



	object 1	integer 2	string 3	table 4
string	MU	Number	Name	Attributes
1	.Models.TableLeg	8	Leg_A	a1
2	.Models.TableLeg	8	Leg_B	a2
3				

Unpublished work. © 2026 Siemens

SimTalk

MUSelection [SimTalk]

Path [SimTalk] - Source

GenerateAsBatch [SimTalk]

getCurrentOrderTableRow [SimTalk]

unshare [SimTalk]

See also

Produce Parts in a Fixed Sequence One Time Only

Unpublished work. © 2026 Siemens

Random [MU selection]

Produces *MUs* according to the values which you type into the frequency table for the setting **MU Selection > Random**.

Remarks

Type the path to this table into the text box next to **Table**.

You can also click ... and select the name of the table in the dialog **Select Object**. *Plant Simulation* automatically formats the table.

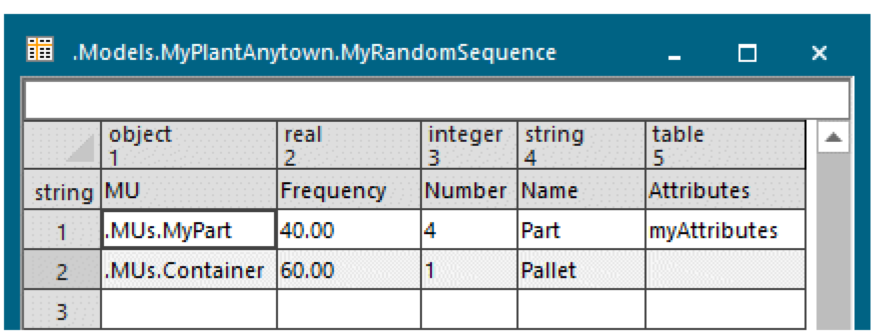
> **Note**
> For the *MUs* to be produced you can also preallocate *user-defined attributes*. If the attribute has the data type *table*, the *MU* gets a reference to the table defined in the *Source*. A reference saves memory and increases performance. In some cases you might need a copy of the table though. You can enforce this by assigning the name `.unshare` to the subtable, compare **unshare**.

Define a frequency number for each type of *MU* in the frequency table. It has five columns with the data types *object, real, integer, string*, and *table*. Each row in the frequency table designates a production order.

- Type the *MU* class into the column **MU**.
- Type the frequency of the respective production order into the column **Frequency**.
- Type the number of *MUs* to be produced into the column **Number**.

You can, but you do not have to, type a **Name** and **Attributes** into the table.

- Type the name of *MUs* to be produced into the column **Name**.
- Type the name of the attribute subtable of the produced *MUs* into the column **Attributes**.


.Models.MyPlantAnytown.MyRandomSequence

	object 1	real 2	integer 3	string 4	table 5
string	MU	Frequency	Number	Name	Attributes
1	.MUs.MyPart	40.00	4	Part	myAttributes
2	.MUs.Container	60.00	1	Pallet	
3					

Unpublished work. © 2026 Siemens

To make the *Source* produce the number of *MUs*, which you typed into the cell **Number** of the **Table** in a single lot, select **Generate as Batch** 

.

If you select **Number Adjustable** as the **time of creation**, the **amount** does not designate the number of *parts*, but the number of *batches* that the *Source* produces. Then the *Source* produces the entire set of parts all at once at the given start time and attempts to move all of the parts on to the next object in a single lot.

To make the *Source* produce the *MUs* as a sequence of individual *MUs*, clear the check box.

To check which part types were produced at which point in time, activate the check box **Creation Table** on the tab **Statistics** and then open the table.

SimTalk

MUSelection [SimTalk]

Path [SimTalk] - Source

GenerateAsBatch [SimTalk]

getCurrentOrderTableRow [SimTalk]

unshare [SimTalk]

See also

Produce Parts With a Random Frequency Entered into a Data Table

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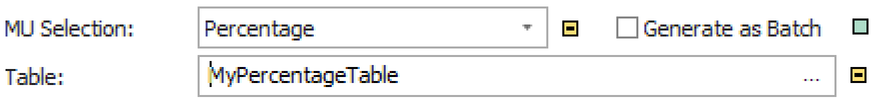
Percentage [MU selection]

Produces *MUs* according to the values which you typed into the percentage table for the setting **MU Selection > Percentage**.



Remarks

Type the path to this table into the text box next to **Table**.

You can also click … and select the name of the table in the dialog **Select Object**. *Plant Simulation* automatically formats the table.

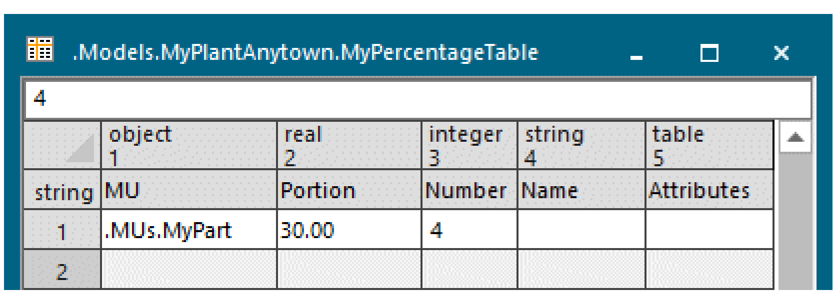
> **Note**
> For the *MUs* to be produced you can also pre-allocate *user-defined attributes*. If the attribute has the data type *table*, the *MU* gets a reference to the table defined in the *Source*. A reference saves memory and increases performance. In some cases you might need a copy of the table though. You can enforce this by assigning the name `.unshare` to the subtable, compare **unshare**.

Define the percentage for each type of *MU* in the percentage table. Each row in the percentage table pertains to a production order.

- Type the name of and the path to the *MU* class into the column **MU**.
- Type the percentage of the respective part types to be produced into the column **Portion**.
- Type the amount of the respective part types to be produced into the column **Number**.

You can, but you do not have to, type a **Name** and **Attributes** into the table.

- Type the name of the respective part types to be produced into the column **Name**.
- Type the name of the attribute subtable of the respective part types to be produced into the column **Attributes**.



Unpublished work. © 2026 Siemens

To make the *Source* produce the number of *MUs*, which you typed into the cell **Number** of the **Table** in a single lot, select **Generate as Batch** ☑ Generate as Batch.

If you select **Number Adjustable** as the **Time of Creation**, the amount does not designate the number of *parts*, but the number of *batches* that the *Source* produces. Then the *Source* produces the entire set of parts at the point in time you entered as **Start time**.

To make the *Source* produce the *MUs* as a sequence of individual *MUs*, clear the check box.

> **Note**
>
> **MU Selection > Percentage** designates a deterministic process for realizing the frequencies of the part types which you typed into the column **Portion** for the part types. When selecting the next part, the *Source* goes to the row on the table which has the greatest need in the column **Portion** at this point in time. This creates periodically repeating patterns in the sequence of the part types to be produced.

SimTalk

MUSelection [SimTalk]

CreationTableActive [SimTalk]

GenerateAsBatch [SimTalk]

getCurrentOrderTableRow [SimTalk]

unshare [SimTalk]

See also

Produce Parts With a Percentage Entered into a Data Table

Exit strategy > **Percentage [described]**

Video on YouTube

https://youtu.be/BvHLQCIjGIA?si=b7Ua-Nl7_uxT39m4&t=158

Unpublished work. © 2026 Siemens

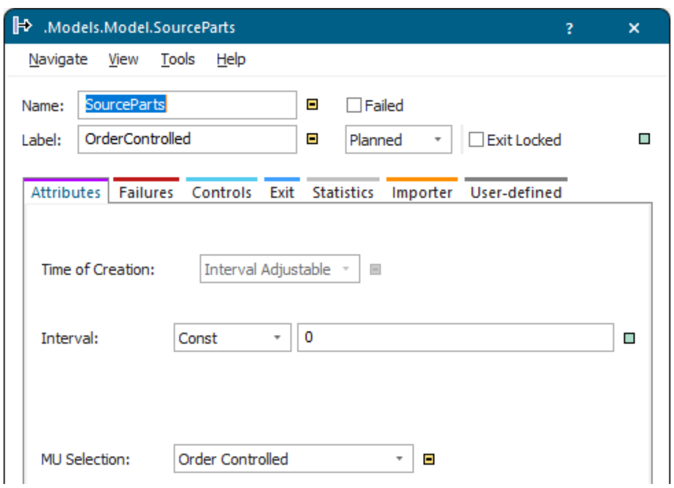
Order Controlled [MU selection]

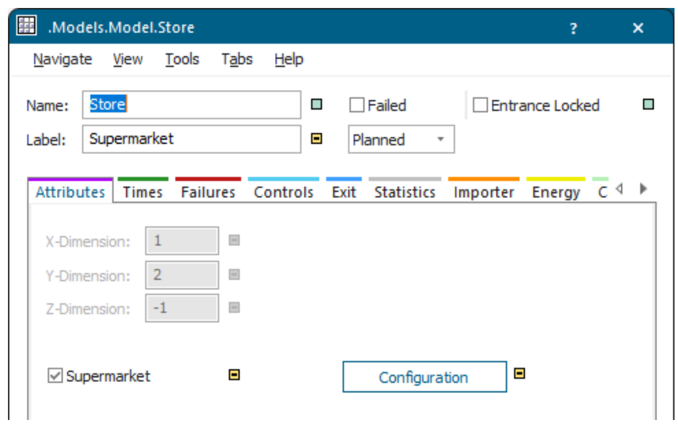
Only produces *MUs* if they are ordered by a *Store* with activated setting **Supermarket** or with the method *orderParts* for the setting **MU Selection > Order Controlled**.

Remarks

Automatic Routing of the produced part delivers it to the ordering station. If **automatic routing** of the part is deactivated, *Plant Simulation* automatically activates it.

The *Source* produces *MUs* according to the setting **Interval Adjustable** and the **Interval** between the creation times, that you type in.





SimTalk

MUSelection [SimTalk]

orderParts [SimTalk] - Source

See also

Store as **Supermarket [check box]**

Configuration [button]

View > **Show Orders [Source]**

Use the Store as Supermarket

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Generate as Batch [check box]

To make the *Source* produce the number of *MUs*, which you typed into the cell **Number** of the **Table** in a single lot, select this check box.

Remarks

If you activate **Generate as Batch**, the *Source* produces the entire set of parts all at once at the given start time and attempts to move all of the parts on to the next object in a single lot instead of one by one.

To make the *Source* produce the *MUs* as a sequence of individual parts, clear the check box.

The setting applies for the **MU Selection** > **Sequence**, **Sequence Cyclical**, **Random**, and **Percentage**.

> **Note**
>
> If you select **Number Adjustable** as the **Time of Creation**, the **Amount** does not designate the number of *parts*, but the number of *batches* that the *Source* produces. Then the *Source* produces the entire set of parts at the point in time you entered as **Start Time**.

SimTalk

GenerateAsBatch [SimTalk]

CurrentBatchNumber [SimTalk]

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Tab Failures

Define **failures** as described under the **Tab Failures**.

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Tab Controls

Provides controls to modify the built-in behavior of the object.

Select the Path to an Existing Method

Click the ellipsis button. Navigate to the location of the *Method* in the dialog **Select Object [for controls]** and click **OK**. This inserts the name of the *Method* into the text box of the **Control**.

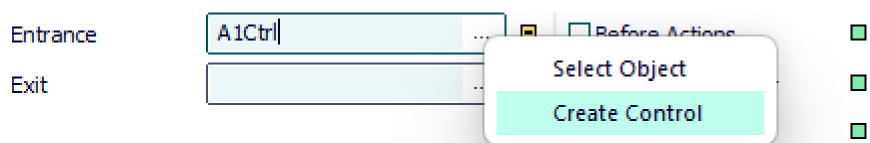
Press **F2** in the text box to open the *Method*. Then type in the source code of the **Control**.

Instead of choosing **Select Object**, you can also select the *Method* in a *Frame*, drag it to the text box and drop it there.

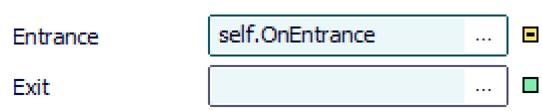
Create a Control That is a Method of the Object

Proceed as follows to create a control as a *user-defined attribute* of data type *Method*:

- Type a meaningful name into the text box and select **Create Control [context menu]**. *Plant Simulation* then inserts `self.Name_you_typed_in_for_the_control`, such as `self.A1Ctrl`.



- Select **Create Control** on the empty text box. *Plant Simulation* then inserts `self.OnBuilt_in_name_of_the_control`, such as `self.OnEntrance`.



  Type the source code of this control into the *Method* that opens.

  To edit the source code later on:

  - Press **F2**.
  - Or hold down **Shift** and double-click into the text box.
  - Or select **Open Object** on the context menu.
  - Or click the tab **User-defined** and double-click the name of the *Method* in the list.

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- To delete this control, delete the *user-defined attribute*. If you only delete the name from the text box, the *user-defined attribute* is retained.

See also

Select Object [for controls]

Entrance Control [general description]

Exit Control [general description]

Shift Calendar [tab Controls]

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Tab Exit

Select to which of its successors the object moves the *MU* on the **Tab Exit**.

See also

Blocking [exit strategy]

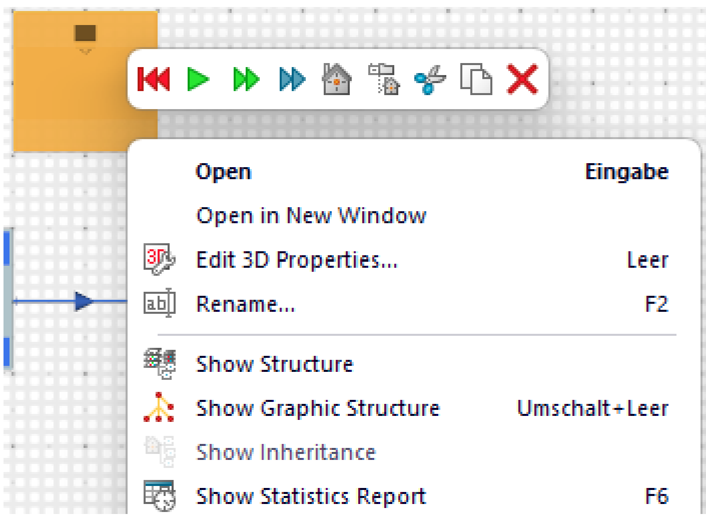
Strategy [material flow objects]

Tab Statistics

Tab Statistics [Source]

In addition to the values described under the **Tab Statistics**, the tab **Statistics** provides the dialog items **Creation Table** and **Open**.

To view **Resource Statistics of Stationary Resources** in the **Statistics Report**, select **View > Show Statistics Report** in the dialog of the object. You can also click the right mouse button in the *Frame* and select **Show Statistics Report** or you can press **F6**.



See also

View and Visualize Statistics

Statistics Report [described]

Resource Statistics [check box]

Resource Type

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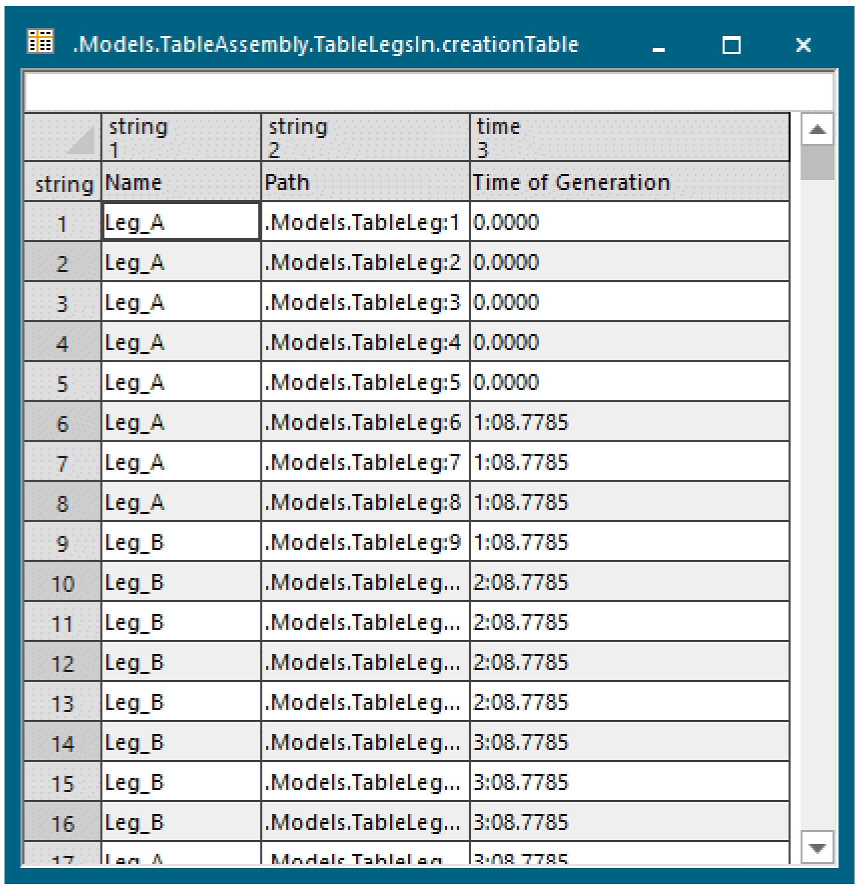
Creation Table [Source]

To write all events, for which the *Source* produced *MUs* during a simulation run, into a table, select this check box.

Remarks

Then click **Open** to open the creation table and check the values.



	string 1	string 2	time 3
string	Name	Path	Time of Generation
1	Leg_A	.Models.TableLeg:1	0.0000
2	Leg_A	.Models.TableLeg:2	0.0000
3	Leg_A	.Models.TableLeg:3	0.0000
4	Leg_A	.Models.TableLeg:4	0.0000
5	Leg_A	.Models.TableLeg:5	0.0000
6	Leg_A	.Models.TableLeg:6	1:08.7785
7	Leg_A	.Models.TableLeg:7	1:08.7785
8	Leg_A	.Models.TableLeg:8	1:08.7785
9	Leg_B	.Models.TableLeg:9	1:08.7785
10	Leg_B	.Models.TableLeg...	2:08.7785
11	Leg_B	.Models.TableLeg...	2:08.7785
12	Leg_B	.Models.TableLeg...	2:08.7785
13	Leg_B	.Models.TableLeg...	2:08.7785
14	Leg_B	.Models.TableLeg...	3:08.7785
15	Leg_B	.Models.TableLeg...	3:08.7785
16	Leg_B	.Models.TableLeg...	3:08.7785
17	Leg_A	.Models.TableLeg...	3:08.7785

> **Note**
>
> The *Source* records the creation events in addition to the resource statistics it collects in any case.

SimTalk

CreationTableActive [SimTalk]

creationTable [SimTalk] - Source

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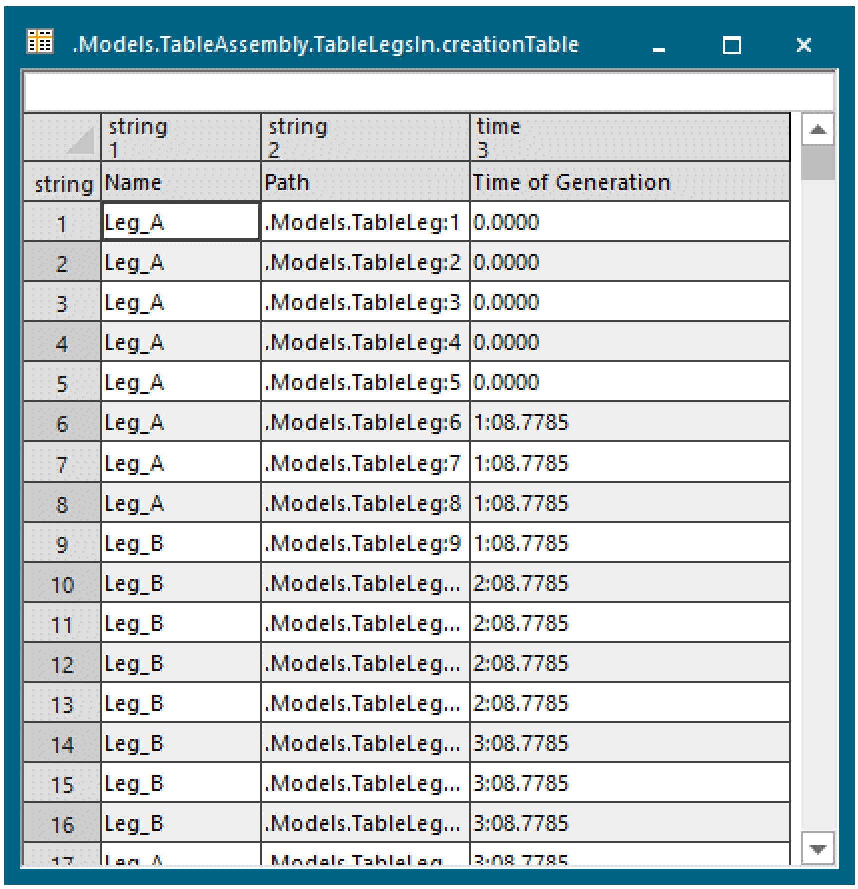
Open [creation table]

To open the creation table of the *Source*, if you selected the check box **Creation Table**, click **Open**.

Remarks

Each row lists a production event:

- The column **Name** shows the name of the *MU* class.
- The column **Path** shows the path of the *MU* class, the name of the *MU*, and its number.
- The column **Time of Generation** shows the time at which the *Source* created the individual *MU*.


.Models.TableAssembly.TableLegsIn.creationTable

	string 1	string 2	time 3
string	Name	Path	Time of Generation
1	Leg_A	.Models.TableLeg:1	0.0000
2	Leg_A	.Models.TableLeg:2	0.0000
3	Leg_A	.Models.TableLeg:3	0.0000
4	Leg_A	.Models.TableLeg:4	0.0000
5	Leg_A	.Models.TableLeg:5	0.0000
6	Leg_A	.Models.TableLeg:6	1:08.7785
7	Leg_A	.Models.TableLeg:7	1:08.7785
8	Leg_A	.Models.TableLeg:8	1:08.7785
9	Leg_B	.Models.TableLeg:9	1:08.7785
10	Leg_B	.Models.TableLeg...	2:08.7785
11	Leg_B	.Models.TableLeg...	2:08.7785
12	Leg_B	.Models.TableLeg...	2:08.7785
13	Leg_B	.Models.TableLeg...	2:08.7785
14	Leg_B	.Models.TableLeg...	3:08.7785
15	Leg_B	.Models.TableLeg...	3:08.7785
16	Leg_B	.Models.TableLeg...	3:08.7785
17	Leg_A	.Models.TableLeg...	3:08.7785

SimTalk

creationTable [SimTalk] - Source

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See also

Creation Table [Source]

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Tab User-defined

Define your own attributes as described under the **Tab User-defined**.

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Navigate Menu

The commands are described under the **Navigate Menu**.

View Menu

View Menu

The **View Menu** provides commands to access its functions.

Refresh [on View menu]	**Forward Blocking List**
Show Statistics Report [on View menu]	**Exit Blocking List**
Show Attributes and Methods [on View menu]	**Associated Lockout Zones**
Show Orders [Source]	**Associated Shift Calendar**
Contents [material flow objects]	

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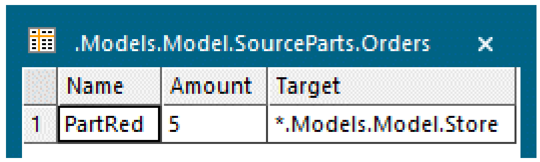
Show Orders [Source]

Opens a table that shows the orders which are pending with the *Source*.

Remarks

The table contains the **Name** of the ordered *part*, its **Amount**, and the **Target** of the *part*. The **Target** is the object that orders the *parts*.



.Models.Model.SourceParts.Orders

	Name	Amount	Target
1	PartRed	5	*.Models.Model.Store

See also

MU selection [drop-down list] > **Order Controlled [MU selection]**

Supermarket [check box]

Configuration [button]

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Tools Menu

The commands are described under the **Tools Menu**.

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Help Menu

The commands are described under the **Help Menu**.

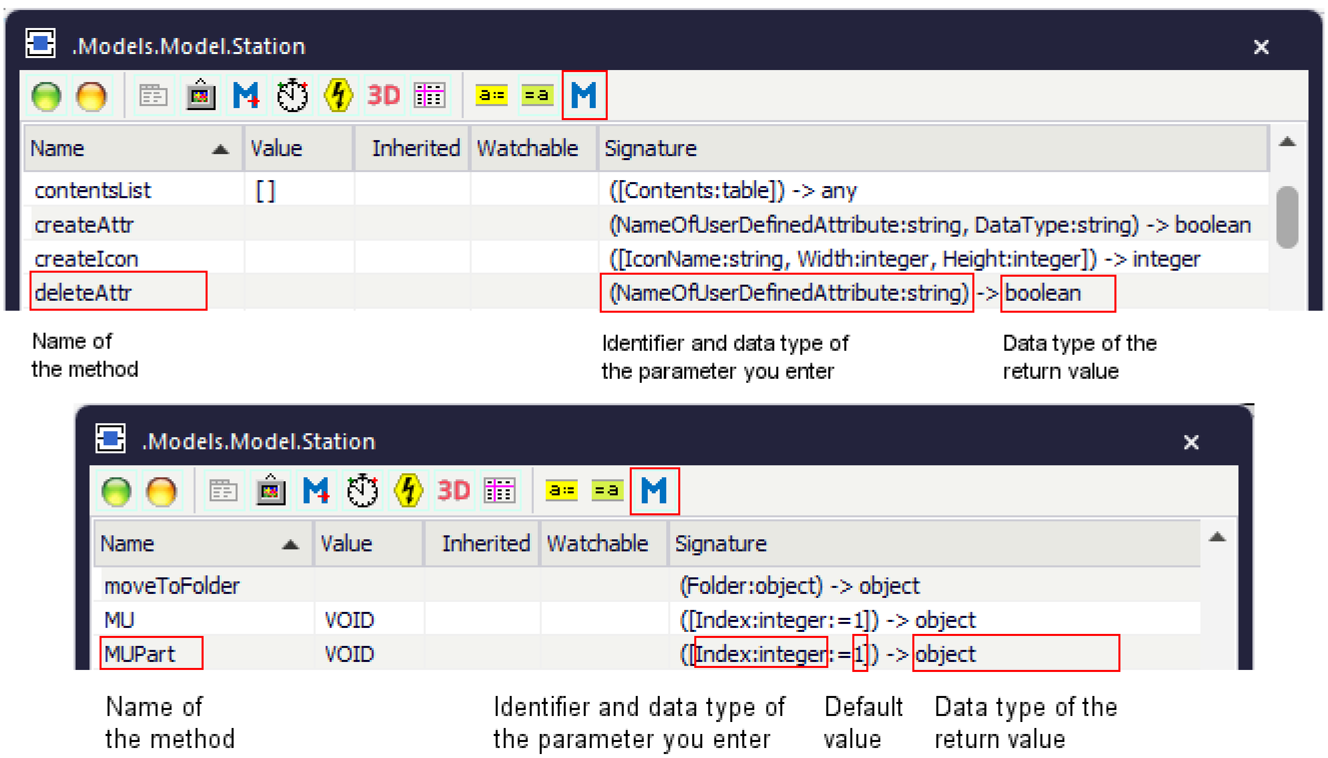
Methods of the Source

_Methods of the Source

The *Source* provides:

- The *methods* listed in the table of contents to the left.
- The **Methods of the Material Flow Objects**.
- The **Methods of All Objects**.

To view all of the *methods*, *read-only attributes*, and *attributes* of the object, open the window **Show Attributes and Methods**. The figure below illustrates the information using the example of the object *Station*.



- Select **Show Attributes and Methods** on the context menu of the **Class Library** to show the *methods*, *read-only attributes*, and *attributes* of the selected **Class [general description]**.
- Press the **F8** key or click **Show Attributes and Methods** on the **Home** ribbon tab of the *Frame* into which you inserted an instance to show the *methods*, *read-only attributes*, and *attributes* of the selected **Instance [general description]**.

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